

RIO DE JANEIRO GALEAO, Brazil

WMO: 837460

Lat: 22.809S

Lon: 43.244W

Elev: 9

StdP: 101.22

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	15.9	10.8	8.1	21.9	12.0	8.7	21.6	8.7	24.4	7.5	23.3	1.5	350	0.299

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.6	36.1	24.6	34.9	24.7	33.8	24.7	27.7	31.4	27.2	30.8	26.6	30.3	3.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.0	22.7	29.9	26.2	21.6	29.1	25.9	21.2	28.8	88.5	32.0	86.0	30.9	83.7	30.5	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.2	6.3	DB	12.7	39.0	1.4	1.3	11.6	39.9	10.8	40.6	10.0	41.3	8.9	42.2	(4)
(5)			WB	11.8	28.7	1.4	1.5	10.8	29.7	10.0	30.6	9.2	31.4	8.2	32.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.7	27.8	28.3	27.1	25.7	23.1	22.1	21.6	22.2	23.1	24.3	25.1	26.9
	DBStd	3.26	2.43	2.10	2.06	2.15	2.07	2.05	2.22	2.33	2.62	2.67	2.78	2.47
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	6	0	0	0	0	0	1	3	1	1	0	0	0
	CDD10.0	5383	552	511	531	470	406	362	361	379	393	444	452	523
	CDD18.3	2348	294	278	272	220	148	113	105	122	143	186	202	264
	CDH23.3	18524	3017	2949	2482	1681	728	499	507	664	838	1223	1513	2423
	CDH26.7	6732	1282	1266	890	511	171	114	136	209	289	402	529	932
Wind	WSAvg	2.7	2.9	2.9	2.7	2.6	2.3	2.2	2.3	2.6	3.0	3.2	3.1	3.0
Precipitation	PrecAvg	1228	177	131	154	109	71	45	44	43	69	94	145	158
	PrecMax	1773	355	278	366	282	129	84	103	124	119	253	328	464
	PrecMin	468	38	30	41	19	27	9	5	16	23	21	15	38
	PrecStd	342	77	69	97	73	30	18	26	27	30	62	75	92
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.1	37.9	36.1	34.9	32.8	31.5	31.9	33.9	35.2	35.2	36.1	37.1
		MCWB	25.0	24.6	24.8	24.7	22.9	21.6	21.0	21.1	21.9	23.4	23.9	24.9
	2%	DB	35.2	35.8	34.1	32.9	30.2	29.8	30.0	31.1	32.2	32.9	33.9	34.9
		MCWB	25.2	25.0	25.3	24.7	22.8	21.6	21.0	20.8	22.0	23.3	24.5	25.1
	5%	DB	33.9	34.1	32.2	31.0	28.2	27.9	28.1	29.0	30.0	30.8	31.8	33.0
		MCWB	25.2	25.2	25.3	24.8	22.4	21.4	20.7	20.9	21.7	23.3	24.1	24.9
	10%	DB	32.2	32.8	31.0	29.4	26.9	26.1	26.1	26.9	27.9	28.9	29.8	31.2
		MCWB	25.3	25.3	25.4	24.3	22.1	21.2	20.5	20.9	21.7	22.8	24.0	25.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.5	28.8	28.2	27.4	25.7	24.2	23.6	24.2	24.7	26.1	26.9	27.7
		MCDB	32.3	33.9	31.5	30.4	28.8	27.0	27.6	28.1	29.6	29.9	31.2	31.7
	2%	WB	27.6	27.7	27.2	26.5	24.6	23.1	22.6	23.2	23.6	25.1	26.0	26.7
		MCDB	31.2	31.5	30.7	29.5	27.6	26.2	26.2	26.7	28.4	29.2	30.2	30.8
	5%	WB	26.9	27.1	26.5	25.7	23.7	22.4	21.8	22.3	22.9	24.3	25.2	26.1
		MCDB	30.6	30.8	30.0	28.7	26.5	25.5	25.3	25.8	26.8	28.3	29.2	30.2
	10%	WB	26.3	26.4	25.9	25.0	22.8	21.8	21.1	21.5	22.2	23.6	24.5	25.5
		MCDB	29.9	30.1	29.2	28.1	25.7	24.9	24.5	25.0	26.0	27.3	28.3	29.5
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.6	7.7	7.7	7.8	8.4	8.8	9.0	8.3	7.6	7.3	7.7
		MCDBR	10.7	11.1	10.0	9.8	9.9	11.1	12.5	13.2	13.0	11.4	11.2	10.8
		MCWBR	3.5	3.5	3.6	3.6	4.1	4.6	5.1	4.9	4.4	3.9	3.8	3.5
	5% WB	MCDBR	8.7	9.2	8.5	8.2	8.4	9.1	9.8	10.1	9.8	9.2	9.0	8.9
	MCWBR	3.9	4.0	4.0	3.7	4.1	4.5	4.9	4.9	4.3	4.1	3.9	3.8	
Clear-Sky Solar Irradiance	taub	0.414	0.411	0.410	0.398	0.382	0.379	0.381	0.413	0.492	0.470	0.435	0.426	
	taud	2.465	2.457	2.467	2.487	2.505	2.501	2.476	2.368	2.123	2.247	2.377	2.420	
	Ebn at Noon	932	921	895	863	836	817	828	835	806	858	907	922	
	Edh at Noon	120	119	112	102	93	89	94	113	156	145	130	126	
All-Sky Solar Radiation	RadAvg	5.98	6.22	5.30	4.66	3.71	3.48	3.60	4.26	4.58	5.13	5.25	5.81	
	RadStd	0.89	0.65	0.45	0.33	0.20	0.26	0.30	0.32	0.58	0.61	0.44	0.54	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	-1.69	N/A	-1.40	-1.66	N/A	N/A	N/A	N/A
Regional (4 neighbors)		N/A	N/A	-1.69	N/A	-1.40	-1.66	N/A	N/A	N/A	N/A

Nomenclature: See separate page